

BRUsens – DSS V9 Strain Sensing Cable

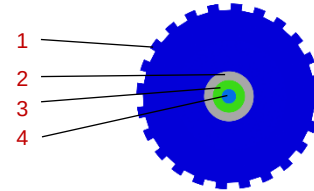
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Distributed Fiber-Optic Strain Sensing (DFOSS) cable with a Fiber in Metal Tube (FIMT) encapsulated optical fiber, and a PA outer sheath, for strain sensing up to 1% (10000 µstrain)

Construction

- 1) PA outer sheath, with structured surface
- 2) Metal tube SS316L for protection and hermetic seal
- 3) Multi-layer buffer and strain transfer layer with interlocking system
- 4) Special strain sensing single mode fiber

BSST V9



Description

- Central hermetically-sealed metal tube with one optical fiber
- High strain sensitivity
- Compact design, good flexibility, small bending radius
- Robust, abrasion resistant sheath, optimized strain transfer
- Longitudinally and laterally watertight
- Good rodent protection
- Halogen-free cable sheath

Temperature range

Operating temperature: -30 °C ... +70 °C
 Storage temperature: -30 °C ... +70 °C
 Installation temperature: -5 °C ... +50 °C

Jacket color

Blue, similar to RAL 5005

Standards

- Cable tests complying with IEC 60794-1-2

Application

- Brillouin Distributed Strain Sensing (DSS)
- Structural Health Monitoring (SHM)
- Pipeline monitoring
- Soil movement, ground monitoring
- Precision measurement and alarm systems
- Direct burial in soil, concrete, composite structures
- Harsh environment, subsea, outdoors

Customization options and services:

- Accessories such as loops, fan-outs, connectors, mounting brackets, anchors etc. available
- Splicing kit for strain sensing cables
- Custom fiber types

Standard Optical fiber

Single mode fiber

Technical data at 20°C:

Type	Cable ø mm	Weight kg/km	Max. tensile strength Installation N	Typical load at 1% elongation N
1F	3.2	10.5	260	470

Type	Min. bending radius		Max. crush res. N/cm
	With tensile	Without tensile	
1F	20xD	15xD	250

Optical fiber data (cabled) at 20°C:

Fiber Type	Attenuation, dB/km		
	850 nm	1300/1310 nm	1550 nm
SMF			≤ 0.5

Fiber Optic Sensing Cable

Typical Brillouin parameters BOTDR or BOTDA at 1550nm

	Temperature sensitivity df_B / dT	Strain sensitivity $df_B / d\epsilon$	Central Brillouin frequency
SMF	$\sim 2 \text{ MHz/}^\circ\text{C}$	$\sim 450 \text{ MHz/\%}$	$\sim 10.7 \text{ GHz}$

Typical Rayleigh parameters at 1550nm

	Temperature sensitivity df_B / dT	Strain sensitivity $df_B / d\epsilon$
SMF	$\sim -4 \text{ GHz/}^\circ\text{C}$	$\sim -0.13 \text{ GHz/\mustrain}$